¿Es fraude? — Judges' brief

A second opinion before you fall for a scam

A free, open-source web app for Mexico, in Spanish and English. It checks a suspicious message, screenshot, link, email, call or loan app against official Mexican sources, quotes every warning sign, never says "safe", and walks victims through reporting it. Built for the Nebius x NVIDIA Global AI Hackathon, Best Apps and Agents track.

Live demo (English)	**Code (MIT)**	**Demo video (2:14)**
es-fraude-seven.vercel.app/?lang=en	github.com/ramsesavi94-maker/es-fraude	youtu.be/5_hwRbQ4zE

Try it in two minutes (no account needed)

1. Open the English demo and tap **Fake tax authority (SAT)**. In about 6–9 s you get a high-risk verdict with quoted signs, the lookalike domain check and real official SAT alerts found by Tavily.
2. Scroll to **Report it, step by step**, tap **Yes**, choose **Ciudad de México**: verified urgent steps, a draft complaint built only in your browser, and one-tap routes (online prosecutor complaint, cyber police, 089 with a script).
3. Tap **Email "from SAT"** (a Gmail account posing as the tax authority) and **Real bank alert** (a legitimate message: no alarm, but never "safe").
4. Optional: the **Screenshot** tab reads a photo of an SMS with Nemotron 3.5 Super VL. A sample is in the repo: `samples/sms-falso-banco.png`.

Example messages are in Spanish, as people in Mexico receive them; with `?lang=en` the analysis is written in English while quotes stay verbatim. The public demo has a per-IP hourly limit and a daily cap to stay within free credits.

Why it matters

Fraud is the most frequent crime in Mexico: an estimated 8.1 million frauds in 2025, 80.6% online or electronic, and 96.5% never reported (UNODC-INEGI, ENVIPE 2026). Victims ask Facebook groups before official channels. ¿Es fraude? answers about the message in front of them and ends in a concrete, verified next step.

How NVIDIA Nemotron and Nebius Token Factory are used

1 · Read	**2 · Verify**	**3 · Reason**	**4 · Act**
Nemotron 3.5 Super VL Nemotron 3 Super Screenshot transcription; extraction of sender, links, phones, amounts, requests.	Deterministic hostname and lookalike checks, domain age (RDAP), Tavily searches limited to official Mexican domains.	Nemotron 3 Ultra Verdict against 12 scam patterns documented by Mexican authorities. JSON mode, temperature 0.	Steps, phone numbers, deadlines and reporting routes come only from a verified catalog.

All inference runs at request time on Nebius Token Factory (OpenAI-compatible API). We compared models on real calls: Super was faster (≈2 s vs ≈3.6 s) and more accurate than Nano for extraction, so we use Super there and reserve Ultra for reasoning. Measured cost: about US$0.017 per analysis.

Results (synthetic sets written from official alerts)

SET	CASES	SCAMS CAUGHT	LEGITIMATE, NO ALARM	NOTES
Third set	40	**25/25**	**15/15**	**Unseen, measured once with no changes**; 27 hard cases; also 25/25 and 15/15 with English answers
Links & emails	30	**18/18**	**12/12**	**Unseen**: lookalike domains, deceptive subdomains, Cyrillic letters, IP hosts, shorteners, free accounts posing as SAT/IMSS/BBVA
Holdout	40	**25/25**	**15/15**	First run before fixes: 24/25 and 14/15
Tuning set	45	**30/30**	**15/15**	Used to tune rules

Run any set with `npx tsx --env-file=.env scripts/evaluate.ts eval/third.json`. The sets are synthetic; testing with real users is the next step.

Safety and grounding decisions

- **No invented contacts.** Early on, Ultra invented a SAT phone number and a non-existent reporting short code. The model now only classifies and explains.
- **Never "safe".** Enforced in code with a fixed message when no signs are found.
- **Quotes are evidence.** Signals must quote the message or an official page; quotes of our own check text are dropped in code.
- **Relevant evidence only.** Search results must name the institution as a whole word and talk about fraud.
- **Privacy.** Nothing is stored; searches receive names, domains and phone numbers, never the message; complaint details stay in the browser.
- **Honest routes.** Each reporting channel says whether it is guidance, a claim or a formal criminal complaint.

Where to look in the repo

`src/analyze.ts`	Pipeline: read, extract, verify, reason, compose
`src/catalog.ts`	Verified patterns, channels, domains
`src/report.ts`	Reporting routes, urgent steps, evidence
`src/linkcheck.ts`	Link and email checks, RDAP
`src/llm.ts`	Token Factory client
`eval/`	Evaluation sets and results
`docs/research/`	Sourced research behind the catalog and routes

Known limitations

- Evaluation is synthetic; real-world accuracy is not yet measured.
- Domain age is unavailable for .mx (no public RDAP).
- Online complaint routes verified for CDMX, State of Mexico and Nuevo León; other states get in-person guidance and the official cyber-police directory.
- In the video, analysis waits are shortened and the narration is AI-generated.

Team: Nelson Ramses Aviles and Aldo Montes de Oca, with Claude Code (architecture and code) and Codex (research on Mexican scam patterns, official reporting routes and evaluation sets). This app is guidance, not legal advice.